Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Lidar for Marine Resource Exploration**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Resources	Resource Survey & Exploration	Shipbuilding/ Marine Systems	Marine Leisure and Exploration Equipment
Concept of Proposed Technology	The underwater lidar system is under development for bathymetric surveys enabling 3D seafloor mapping, and infrastructure monitoring (e.g. offshore wind facilities). Seabed morphology can be used to identify areas for probable resource deposits. The lidar can be operated from a remotely operated vehicle and collects point cloud data in real-time.			
Rationale for Joint R&D Project	CSEM has developed the lidar hardware and system development and integration know-how, however several opportunities exist for furthering the technology and tailoring it to specific applications. Collaboration with a Korean SME with competence in underwater vehicle navigation or marine inspection would enable mutually beneficial lidar advancement for use in seabed mapping.			
Key Words	1. Bathymetric survey 2. Seafloor mapping and morphology 3. Underwater simultaneous localization and navigation 4. Resource deposit site identification 5. Underwater infrastructure inspection			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 5	End of the Project	Level 6
R&D Objectives	Enable real-time seabed mapping for terrain morphology reconstruction, ROV navigation and identification of resource deposit sites. Advance lidar technology for marine resource exploration. Leverage bathymetric survey for novel approaches to marine resource identification.			
R&D Target Specifications	• Real-time 3D point cloud output suitable for closed-loop simultaneous localization and navigation. • Robust operation in turbid conditions • Operational range: up to tens of meters in water • Frame rate / acquisition rate: compatible with vehicle dynamics and motion to support real-time data processing • Dual wavelength lidar for chemical properties identification of resources			
R&D Plan	Phase 1 – Requirements Definition: Concept of operations and use case definition; definition of interfaces with onboard systems; lidar specifications. Phase 2 – Technology Development and Prototyping: Implementation and optimization of lidar system for underwater bathymetry; signal processing and algorithms development for terrain reconstruction, hazard detection, and navigation support. Phase 3 – Test and Iteration: Functional test and preliminary system integration and test combining lidar hardware with vehicle interfaces and algorithms. Software iteration based on preliminary algorithm performance. Phase 3 – Validation: Closed-loop demonstrations with lidar for seabed morphology mapping and inspection.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lidar specifications and overall system architecture definition based on use cases and concept of operations • Lidar system expertise and interface definition • Lidar functional and performance test • Support to lidar integration and validation
Expected R&R of Korean SME Partner	• Definition of use cases and concept of operations • Definition processing and computing requirements aligned marine resource exploration objectives • Development and optimization of processing hardware and/or algorithms • Lidar integration and validation activities

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Outcome**: TRL advancement and integration of lidar system in marine vehicle for marine resource exploration. **Impact:** The proposed development of a lidar for marine resource exploration will enable safe autonomous navigation of marine vehicles as well as 3D mapping of seabed morphology. The depth sensing capabilities of lidar combined with point cloud processing algorithms for localization and mapping provide terrain information which allows for the identification of probable marine resource deposit locations. The terrain mapping can be used to narrow search areas for further investigation with other sensing technologies. The lidar data can be further fused with other sensor data to develop situational awareness of the marine environment, providing detail context of an area of interest.

6. References

(1) Bosch, E., Langenegger, F., Gege, P., Roulet, J. C., Nguyen, D., Holzer, J., ... & Pache, C. (2025). Preserving underwater cultural heritage: combining satellite images with a novel flash lidar platform. *Italian Journal of Engineering Geology and Environment*, 25-33.

(2) Pache, Christophe, et al. "Miniature flash LiDAR for bathymetry: an experimental proof-of-concept." 2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS. DOI: 10.1109/IGARSS47720.2021.9554034

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization